MILITARY SPECIFICATION

HEATER; AIRCRAFT, COMBUSTION TYPE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the requirements for a combustion type aircraft heater and the accompanying ignition system which includes either a spark plug or glow plug igniter, and electrical accessories to operate the igniter. In general, hereinafter, heater is used in lieu of heater and ignition system.

1.2 Classification. This specification defines requirements for heaters with heat output ratings of 15,000 to 100,000 British Thermal Units (BTU) per hour and every 100,000 BTU thereafter.

1.2.1 The required heater performance characteristics shall be based on the individual airframe manufacturer's curve(s) for the specific aircraft in which the heater is to be used (see figure 1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ–P–416	—Plating: Cadmium, Electrodeposited
QQ–S–763	—Steel Bars; Shapes and Forgings—Corrosion Resisting
QQ–Z–325	—Zinc Coating; Electrodeposited, Requirements for

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MIL–P–116	—Preservation; Methods of
MIL–D–3945	—Detector Kit, Carbon Monoxide, Colormetric
MIL–T–3948	—Tube, NBS Colormetric Indicating, Carbon Monoxide
MIL–C–5015	—Connectors; Electrical
MIL–J–5161	—Jet Fuel, Referee
MIL–E–5272	—Environmental Testing; Aeronautical and Associated Equipment, General Specification for
MIL–E–5557	—Enamel; Heat-Resisting, Glyceryl-Phthalate, Black
MIL–G–5572	—Gasoline; Aviation, Grades 80/87, 91/96, 100/130, 115/145
MIL–J–5624	—Jet Fuel, Grades JP–3, JP–4 and JP–5
MIL–T–5842	—Transparent Areas, Anti-icing, Defrosting and Defogging Systems, General Specifications for
MIL–C–6021	—Castings; Classification and Inspection of, for Aeronautical Application
MIL–I–6181	—Interference Control Requirements; Aircraft Equipment

FSC 1660

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MIL-S-6721 —Steel, Corrosion and Heat Resistant (chemically stabilized) Plate, Sheet and Strip
MIL-W-6858 —Welding; Resistance: Aluminum, Steel or Alloys, Magnesium, Non-Hardening Nickel Alloys, Heat-Resisting Alloys and Titanium Alloys; Spot and Seam
MIL-P-6906 —Plates; Information and Identification
MIL-S-8879 —Screw Threads; Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-P-7936 —Parts and Equipment; Aeronautical, Preparation for Delivery
MIL-P-8585 —Primer Coating; Zinc Chromate, Low Moisture Sensitivity
MIL-W-8611 —Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resistant
MIL-A-8625 —Anodic Coatings for Aluminum and Aluminum Alloys
MIL-T-9107 —Test Reports; Preparation of
MIL-A-9482 —Anti-icing Equipment for Aircraft Heated Surface Type, General Specifications for
MIL-H-18325 —Heating and Ventilating Systems, Aircraft, General Specification for
MIL-T-18607 —Thermal Anti-Icing Equipment, Wing and Empennage
MIL-D-70327 —Drawings; Engineering and Associated Lists

STANDARDS

MILITARY

MIL-STD-129 —Marking for Shipment and Storage
MIL-STD-143 —Specifications and Standards; Order of Precedence for the Selection of
MIL-STD-704 —Characteristics and Utilization of Aircraft Electric Power
MS33586 —Metals; Definition of Dissimilar

(Copies of applicable documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEFENSE DOCUMENTATION CENTER FOR SCIENTIFIC AND TECHNICAL INFORMATION

U. S. Standard Atmosphere, 1962

(Application for copies of this publication should be addressed to the Defense Documentation Center for Scientific and Technical Information (DDC), Building 5, Cameron Station, Alexandria, Va. 22314.)

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Power Test Code—Part 2, Pressure Measurement
PTC 19.2.1.6 —1941 Chapters 1 and 6
PTC 19.2.2.3 —1949 Chapters 2 and 3
PTC 19.2.5 —1952 Chapter 5
Power Test Code—Part 5, Measurement of Quantity of Materials
PTC 19.5.4 —1959 Chapter 4

(Application for copies should be addressed to the American Society of Mechanical Engineers Publications, 29 West 39th Street, New York 18, New York.)

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3. REQUIREMENTS

3.1 Preproduction test. The heaters furnished under this specification shall be a product which has been tested and passed the preproduction tests specified herein.

3.2 Rating. Detail and envelope requirements for heaters furnished under this specification are specified in Table I. Detail requirements for all additional heater components, including auxiliary ignition system and blower(s) when required, shall be supplied by the airframe manufacturer.

3.3 Materials. Materials shall conform to applicable specifications and standards as specified herein. Materials which are not covered by applicable specifications or standards or which are not specifically described herein shall be of the best commercial quality, of the lightest practicable weight, and suitable for the purpose intended. The material shall be free from all defects and imperfections that may affect the quality and reliability of the finished product.

3.3.1 *Section of materials.* Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with MIL–STD–143 except as provided in the following paragraphs.

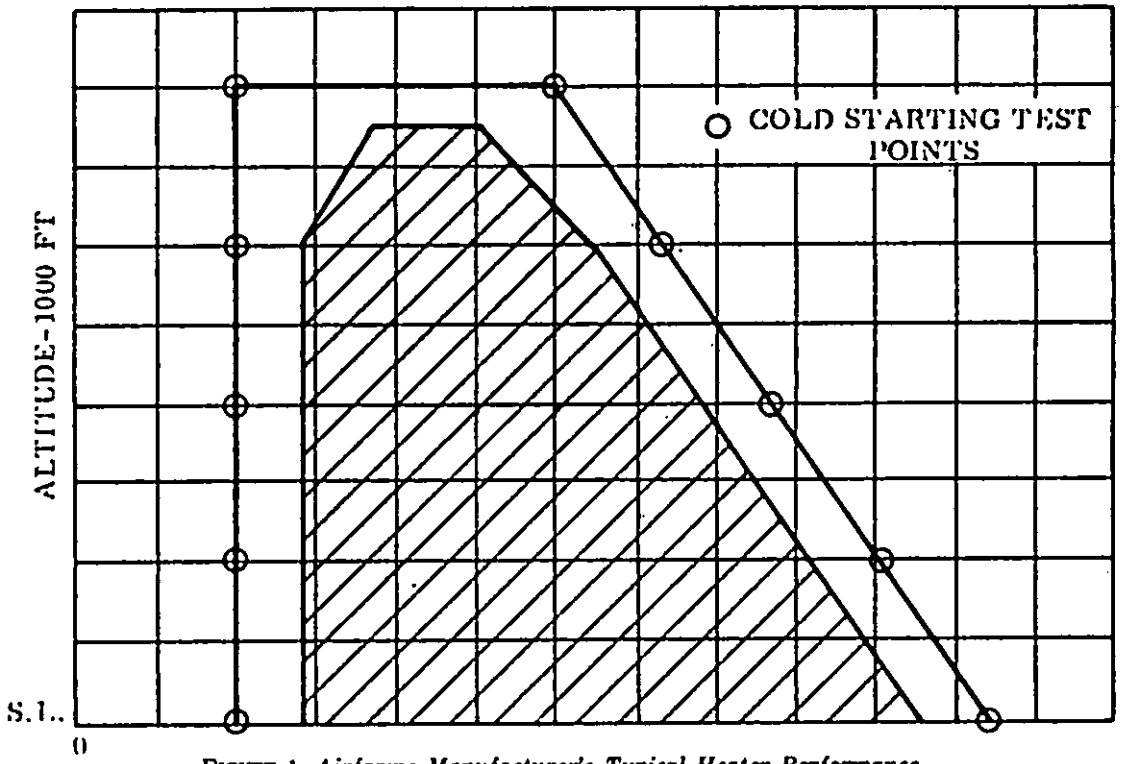


FIGURE 1. *Airframe Manufacturer's Typical Heater Performance*

COMBUSTION CHAMBER DIFFERENTIAL TOTAL AIR PRESSURE—"H_2O
MAXIMUM VENTILATING DIFFERENTIAL TOTAL AIR PRESSURE ACROSS HEATER—"H_2O

NOTE:
1. The cold starting points shall be established at equal altitude increments from sea level to the maximum altitude.
2. The enclosed shaded area indicates rated heat output.
3. Areas for values less than rated heat output shall be indicated.
4. The heater efficiency, based on the high heat value for the fuel shall be specified for the area of rated heat output.

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TABLE I. *Heater ratings, sizes and weights*

Rating BTU Per Hour	*Volumetric Size Cubic Inches (Max.)	Envelope Size (Max.)	**Weight Pounds (Max.)
15,000	700	To be supplied by airframe manufacturer.	20
25,000	1,000		26
50,000	1,000		32
100,000	2,000		38
Further increases in increments of 100,000. BTU per hour.	Increases 1,000 cubic inches per each 100,000 BTU per hour additional rating		Increases 15 lb. per each 100,000 BTU per hour additional rating.

* Excludes combustion air protrusions.
** Includes the ignition system.

3.3.1.1 *Standard parts.* Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by the Standard parts without alteration, and provided the corresponding Standard part numbers are referenced in the parts list and if practicable on the contractor's drawings. In the event there is no suitable corresponding Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3.2 *Metals.* All metals used in construction of heaters shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion and shall not be subject to deleterious changes on exposure to operating temperatures during the normal service life of the heater. The use of dissimilar metals, as defined in MS33586, shall be avoided wherever practicable. Where such contacts are unavoidable they shall be properly insulated in the best practicable manner.

3.3.2.1 *Castings.* Castings shall be of high quality, clean, sound, and free from blow holes, porosity, cracks, and other defects. Castings shall meet the requirements specified for Class 2B of Specification MIL–C–6021.

3.3.3 *Fungus-proof materials.* Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the Government.

3.4 **Design.** The design of the heaters shall be such that combustion can be started by a spark plug or glow plug igniter with the control at a station remote from the heater. Common air inlets for combustion and ventilating air shall not be used. The combustion chamber shall form a closed duct having no openings into the ventilating airstream. The combusion chamber and the inlet and exhaust tubes or ducts shall be so designed that they will not rupture under the most severe explosion conditions that may result from any possible fuel-air mixture. The heater exhaust shall be provided with a leakproof connection to permit attachment of necessary ducting to conduct exhaust gases to the exterior of the aircraft.

3.4.1 *Heat exchanger.* The heater shall be designed so that for any condition of operation, except when positive pressure is imposed at combustion air outlet, with or without a blower, the static pressures existing in any part of the combustion chamber will be less than the static pressure in the ventilating air chamber surrounding the combustion chamber. The heat and corrosion-resistant material used in the construction of

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the heat exchanger and exhaust tubes shall conform to or exceed the requirements of Specification QQ–S–763, Class 321 and MIL–S–6721, and shall be a type which at anytime during the normal service life of the heater can be repaired by Overhaul and Repair Activity procedures.

3.4.2 *Fuel drains.* Fuel drains shall be provided in the combustion chamber to eliminate the possibility of any accumulation of fuel.

3.4.3 *Supply fuel pressure.* The heater shall operate satisfactorily within the fuel-pressure supply limits of 1 to 28 pounds per-square inch for gasoline or more for JP type fuels as specified in the procurement document (see 6.3).

3.4.4 *Vibration.* The heater shall be so designed as to be capable of meeting the vibration requirements of Procedure XII of Specification MIL–E–5272 without damage or improper operation.

3.4.5 *Explosion.* The heater shall be capable of withstanding the explosion resistance test outlined in paragraph 4.8.5 without rupture.

3.5 Construction. The heater shall be so constructed as to withstand the normal strains, shocks, vibrations, and such other conditions as are incident to shipping, storage, installation and service use without failure.

3.5.1 *Interchangeability.* All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL–D–70327.

3.5.2 *Replaceability.* It shall be practicable to replace any component assembly of the heater without requiring replacement of other components.

3.6 Electrical requirements and properties.

3.6.1 *Operating voltage.* The heater shall maintain the specified performance using d-c power (category B) with charncteristics which are listed in Military Standard MIL–STD–704.

3.6.2 *Power Consumption.* The electrical power consumption of the heater when operating under any condition, except during starting, shall not exceed 75 watts. The electrical power required for starting the heater shall not exceed 350 watts.

3.6.3 *Shielding.* The ignition system of the heaters shall utilize high tension shielded type wire and shall meet the radio interference requirements of Specification MIL–I–6181.

3.6.4 *Electrical connectors.* Electrical connectors shall be in accordance with Specification MIL–C–5015.

3.6.5 *Electrical accessories.* The part of the ignition system external to the heater or combustion or ventilation ducts must be incapable of acting as a source of ignition for fuel, oil, or fuel and oil fumes.

3.7 Threads.

3.7.1 *Straight threads.* All straight screw-threads shall conform to Specification MIL–S–8879.

3.7.2 *Pipe-threads.* Pipe threads may be employed only for permanently plugging drilled or cored openings.

3.8 Finish. Protective coatings and finishes that will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.8.1 *Aluminum-alloy parts.* Aluminum-alloy parts, where practicable, shall be covered with an anodic-film conforming to Specification MIL–A–8625, except that electrical receptacles need not be anodized.

3.8.2 *Steel parts.* Steel parts except for corrosion-resisting steel shall be cadmium- or zinc-plated where practicable in accordance with Specification QQ–P–416 or QQ–Z–325, respectively.

3.8.3 *Exposed parts.* Exposed corrosion-resistant material need not be painted. All other exposed metal surfaces shall be coated

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with one coat of zinc chromate primer in accordance with Specification MIL–P–8585, and finished with one coat of black enamel in accordance with Specification MIL–E–5557.

3.9 **Operational performance and product characteristics.** The ignition system of the heater shall be capable of igniting the fuel-air mixture in the combustion chamber at any inlet air and fuel temperature ranging from −54° to 38° C (−65° to + 100° F), except when grade JP–5 fuel is specified, and at any combination of altitude differential pressures as determined applicable for the type of aircraft in which the heaters are to be used. The minimum fuel temperature when grade JP–5 fuel is specified shall be that temperature corresponding to a fuel viscosity of not less than 12 centistokes. The time required for ignition shall not exceed 15 seconds for gasoline or 60 seconds for JP type fuels. Ignition time is defined as the time required after actuation of the ignition switch until the heater exhaust temperature rises 150° F providing that the heater then continues to operate in a normal manner; i.e., no smoking, no backfiring, etc. Heaters shall be capable of operating continuously with fuel conforming to Specification MIL–G–5572, Grade 115/145, or MIL–J–5624, as applicable, for a period of not less than 200 hours without the necessity of removing lead compound, gum, soot, or other deposits which might interfere with the satisfactory operation of the heater. Provisions shall be made for ready removal of deposits accumulated at the end of this period unless the design of the heater is such that deposits do not seriously interfere with its operation during the complete endurance test as outlined in paragraph 4.8.6.

3.10 **Product performance.** The heater shall comply with the performance requirements specified in section 4 when subjected to the applicable tests therein.

3.11 **Special tools.** The design shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, or service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.12 **Identification of product.**

3.12.1 *Nameplate.* The nameplate shall conform to Specification MIL–P–6906, and shall be securely attached to the heater and legibly marked with the following minimum information:

HEATER; AIRCRAFT, COMBUSTION TYPE

Specification MIL–H–5484C

Voltage_____

Heater Rating _____BTU per hour

Manufacturer's Part No._____

Manufacturer's Serial No._____

Contract or Order No._____

Manufacturer's name or trademark

U. S. Property

3.13 **Workmanship.** It is required that all details of workmanship be of the highest quality in order to insure proper functioning of the heater. Particular attention shall be given to thoroughness of assembly, alignment of parts and tightness of assembly screws and bolts, marking of parts, surface finishes, and removal of burrs and sharp edges.

3.13.1 *Welding.* Joining of component parts by fusion welding shall be accomplished in accordance with Specification MIL–W–8611. When resistance welding is used, the requirements of Specification MIL–W–6858 shall be adhered to.

3.13.2 *Cleaning.* All parts of the heater assembly shall be clean and free from dirt, metal chips and other foreign matter during and after assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection require-

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ments as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conforming to prescribed requirements.

4.2 Classification of tests. The inspection and testing of heaters shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests

4.3 Preproduction tests. The preproduction tests shall consist of all tests detailed herein, including flight test.

4.3.1 *Sampling instructions.* Sample heaters of the type ordered shall be furnished by the manufacturer for preproduction testing. Preproduction test samples shall consist of two of each size (rating) heaters upon which preproduction approval is required.

4.3.1.1 *Preproduction test at non-government facility.* Unless otherwise specified, all laboratory testing shall be conducted at the manufacturer's plant under supervision of the Government Inspector. Manufacturers not having adequate laboratory facilities shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The manufacturer shall furnish test reports in accordance with Specification MIL–T–9107 showing results of all tests including a detailed statement indicating compliance or extent of non-compliance with each requirement of this specification, referring specifically to the applicable paragraph numbers in this specification.

4.3.1.2 *Preproduction test at a Government facility.* At the option of the procuring agency, tests may be specified at a Government agency. In such cases, the manufacturer shall forward the test samples to the testing agency named, accompanied by one complete set of drawings for each size heater. The drawings shall include a cutaway section showing all parts and their normal assembled position and shall specify part numbers of all parts and sub-assemblies. The following data shall be furnished on or together with the drawings:

a. Mounting dimensions

b. Overall dimensions

c. Location, size and type of connectors

d. Materials and construction, treatment and finish

e. Nominal rating (BTU/hour)

The drawings required shall be specified by the ordering activity in accordance with paragraph 6.2 of Specification MIL–D–70327. Prior to forwarding the test samples, additional samples shall have been tested by the manufacturer or an approved commercial testing laboratory, and a test report as described in paragraph 4.3.1.1 shall be submitted to the Government testing agency named. An identification name plate shall be securely attached to each sample heater and marked with the information described in paragraph 3.12.1.

4.3.1.3 *Waiver of preproduction tests.* Preproduction tests will not be required on subsequent orders from previous suppliers of aircraft combustion heaters, provided the product has completed all test requirements on a previous order, including flight tests, and provided the product or processes have not been changed, and a certified statement to that effect is furnished to the procuring agency. The certified statement shall list the name and part number of each heater involved together with the letter or letters of previous approval.

4.3.2 *Tests.* The preproduction tests of the heaters shall consist of all tests of this specification, and examination necessary to determine suitability of design. Unless otherwise specified, tests of the two heaters shall normally be conducted in the order given for each heater. The normal sequence of tests for one of the heaters is as follows:

Combustion Chamber Leakage

Ventilating Pressure Resistance

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Explosion Resistance (Initial)
Combustion Chamber Leakage
Endurance
Combustion Chamber Leakage
Explosion Resistance (Final)
Combustion Chamber Leakage
Carbon Monoxide Indication
Exchanger Differential Pressure Resistance
Performance
Combustion Chamber Leakage

The normal sequence of tests for the other heater is as follows:

Combustion Chamber Leakage
Orientation
Altitude Acceptance Calibration
Cold Starting
Delayed Ignition
Fuel Pressure Variation
Vibration
Combustion Chamber Leakage
Explosion Test of Ignition System Accessories
Radio Interference
Humidity
Sand and Dust
Fungus Resistance
Salt Spray
Performance
Combustion Chamber Leakage
Flight Test

4.4 Contractor's responsibility. The contractor is responsible for the performance of the examination of product and the combustion chamber leakage tests (see 4.8.1 and 4.8.2) on each heater prior to submission for Government acceptance.

4.5 Quality conformance tests. Quality conformance tests shall be performed under the surveillance of the Government Inspector on lots submitted for acceptance under contract. These tests shall be conducted after the heaters have successfully passed tests required under 4.4. The quality conformance tests of the heater shall consist of Individual tests and Sampling tests of this specification.

4.5.1 *Individual tests.* The individual tests shall consist of the following tests conducted on each heater:

Examination of product
Combustion chamber leakage

4.5.2 *Sampling tests.* Two heaters of the same rating which have passed the Individual tests shall be selected at random by the Inspector from every 100 or less individual heaters of each lot with the same manufacturer's part number, and subjected to the tests specified in 4.8.3 and 4.8.2.1.

4.5.2.1 *Lot.* A lot shall consist of all heaters of the same part number submitted for inspection at the same time and place.

4.6 Acceptance stamp. Each accepted heater will be marked with the Government Inspector's acceptance stamp.

4.7 Rejection and retest. Any heater failing to meet the requirements of the quality conformance tests shall be rejected. When any random test sample fails to meet the requirements of the sampling tests, the lot represented shall be rejected. Heaters which have been rejected may be replaced or repaired to correct the defects and resubmitted for test. When this has been accomplished, all quality conformance tests shall be repeated. Before resubmitting, full particulars concerning previous rejections and the action taken to correct the original defects shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Inspector.

4.8 Test conditions.

4.8.1 *Atmospheric conditions.* Unless otherwise specified, all tests required by this specification shall be made at ambient laboratory temperature and pressure conditions. Altitude tests shall be made at the approximate

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standard atmospheric pressure and temperature conditions indicated in the U. S. Standard Atmosphere, 1962.

4.8.2 *Operating Voltage.* Unless otherwise specified, the operating voltage shall be 28.5 volts d-c for all operational tests conducted on the heater.

4.8.3 *Fuel.* Unless otherwise specified, the fuel for the operational tests conducted on the heater shall be as specified in Table II.

4.8.4 *Test apparatus and instrumentation.*

4.8.4.1 *Temperature measurements.* Temperature measurements may be made with either a mercury-in-glass thermometer, a thermocouple, or an electric resistance thermometer. Measurements shall be made in accordance with the applicable power test code noted in 2.2.

TABLE II. *Fuel schedule for operational tests (see legend)*

Fuel specified to supply the heater when installed in the specific aircraft (see 6.3)	Fuel to be used during	
	All operational tests except cold starting test	Cold starting test (see 4.8.11)
A	A	A
B	D	D
C	E	E
A and B	D	A and D
A and C	E	A and E
B and C	E	D and E

Legend: A—Specification MIL–G–5572, Grade 115/145
B—Specification MIL–J–5624, Grade JP–4
C—Specification MIL–J–5624, Grade JP–5
D—Specification MIL–J–5161, Grade I
E—Specification MIL–J–5161, Grade II

4.8.4.2 *Air-flow measurements.* The air-flow shall be measured by means of standardized nozzles or orifice plates as outlined in the power test code noted in 2.2. The use of other than the primary elements noted in this publication is acceptable, provided such elements have been calibrated against either a standardized nozzle or orifice.

4.8.4.3 *Pressure measurements.* Pressure measurements shall be made in accordance with the applicable power test code noted in 2.2. Unless otherwise specified, the ventilating and combustion air-pressure measurements shall be measured as total pressure by means of impact tubes. To simplify the test installation, the ventilating total pressures for all heaters shall be measured in 6-inch nominal standard pipe or duct of equivalent diameter whenever practicable. The combustion air inlet shall be measured as total pressure in a duct ordinarily the same diameter as the combustion air adapter to the heater or the combustion air regulator, if provided. This measurement shall be taken in a duct longitudinally parallel to the ventilating air flow. Pressure losses downstream of this measurement, incident to changing the direction of flow to introduce the combustion air into the heater, shall be considered as due to the design of the heater. Reasonable precautions, however, such as smooth elbows, with a radius ratio of 1.5 minimum (radius of bend divided by diameter of duct) shall be used in the test procedure to minimize losses from this source. The combustion air outlet pressure shall be measured in a duct the same diameter as the heater exhaust. The differential pressure across the combustion chamber shall include losses incurred in the combustion air regulator or other such devices.

4.8.4.4 *Carbon-monoxide measurement.* The equipment used to measure carbon-monoxide contamination shall conform to Specifications MIL–D–3945 and MIL–T–3948.

4.8.4.5 *Radio-interference measurement.* Conducted and radiated interference shall be measured by means of equipment conforming to Specification MIL–I–6181.

4.9 Test methods.

4.9.1 *Examination of product.* Each heater shall be examined for conformity with en-

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velope and detail requirements of this specification for which no specific tests are described and to determine conformance to manufacturer's approved materials, fabrication, tolerances, specifications and drawings.

4.9.2 *Combustion chamber leakage.* The combustion chamber shall be tested for leaks by subjecting the chamber to an internal air pressure of 5 pounds per square inch after sealing the combustion air inlet and outlet, fuel drains, etc. With the air supply shut off, the pressure within the chamber shall exceed 4.5 pounds per square inch at the end of 5 minutes. Subsequent to the endurance test the pressure within the chamber shall exceed 2.5 pounds per square inch at the end of 5 minutes. There shall be no leaks which could allow liquid fuel to enter the ventilating air stream or aircraft when the heater is mounted in any normal position with drains open. There shall be no resulting damage to any part of the heater.

4.9.3 *Ventilating pressure resistance.* The heater shall be tested for pressure resistance of the ventilating air side by sealing the ventilating air inlet and outlet and maintaining 10 pounds per square inch air pressure in the sealed space for one minute. Leakage shall not exceed 1.5 pounds of air per hour for each 10,000 BTU per hour. There shall be no resulting damage to any part of the heater.

4.9.4 *Orientation.* The heater shall operate satisfactorily for a period of at least 15 minutes in each of the following positions:

90° simulated dive of the heater from a horizontal position.

90° simulated climb of the heater from a horizontal position.

Horizontal position and rotated about its longitudinal axis at 90° intervals.

4.9.5 *Explosion resistance.* With the heater at a temperature within the range of −25° to +70° F, introduce a fuel-air mixture in a ratio of 0.085 to 0.095. Purge the combustion chamber with this mixture to the extent of at least 10 times the volume of the combustion chamber and heat exchanger assembly. Ignite the mixture with the heater ignition system and repeat the procedure for a total of 25 times. The ventilating air flow shall not have changed more than 10 percent at any comparative condition as determined by an isothermal ventilating air flow calibration before the initial, and after the final explosion tests.

4.9.6 *Endurance.* One of the heaters submitted for preproduction tests shall be connected to a suitable source of air, and the initial differential total pressures shall be sufficient for the heater to deliver at least rated heat output. The initial differential total pressures (ventilation and combustion) shall be maintained during all continuous operation, except that the ventilating air differential total pressure shall be reduced to give a minimum of 75 percent average of the rated heat output during rapid cycling. Inlet air temperatures shall be maintained between 7.2° and −31.7° C (45° and −25° F) throughout the test. The heater shall be operated on a cycle of 4 hours ON and ¼ hour OFF for a total period of 200 hours, excluding the ¼ hour OFF period. The accumulated products of combustion in the heater after the 200-hour period shall not be sufficient to cause more than a 5 percent decrease in output. After inspection, the heater may be disassembled as far as practical and cleaned, if necessary, prior to placing the heater in operation for an additional 200 hours. This procedure shall be continued until the heater has been operated for the required number of hours. After each 200-hour period of operation and at the completion of the required endurance hours the heater shall be examined for evidence of failure. At the conclusion of the endurance test, the heater, after cleaning, shall be operated at initial endurance test conditions and the output shall not have decreased more than 5 percent from the initial endurance test output.

4.9.6.1 *Heater endurance details.* The heater shall be operated for a total of 1000 hours. All operation shall be continuous during each 4-hour ON period, except between 500 and

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700 hours of endurance time when a cycling switch shall be set to control the ventilating air maximum outlet temperature. In this period the heater shall be rapidly cycled at a minimum of 75 percent average of the rated heat output. The endurance test sequence is shown in Table III.

4.9.6.2 *Repairs during test.* No maintenance or repairs shall be accomplished during the test except cleaning and igniter replacement. Whenever necessary, due to ignition failure during the course of the endurance test, the igniter may be removed for cleaning, adjusting, or replacing. At the conclusion of 1000 hours a total of not more than four igniter removals shall have occurred.

4.9.7 *Carbon monoxide indication.* A minimum of five ventilating air outlet samples shall be taken at varying pressure, covering the sea level pressure range as determined applicable for the specific aircraft in which the heater is to be used. No carbon monoxide shall be present as a result of heater operation. Measurement equipment shall be as noted in 4.7.4.4.

TABLE III. *Heater endurance test schedule*

Endurance Time	Ventilating Air		Minimum Heat Output	Operating Conditions
	Temperature Rise	Maximum Outlet Temperature		
Hours	°F	°F		
0–500	225 to 250	.....	Rated	Continuous
500–600	.....	250±10	75% Rated	Rapid Cycling
600–700	.....	350±10	75% Rated	Rapid Cycling
700–800	325 to 350	.....	Rated	Continuous
800–1000	225 to 250	.....	Rated	Continuous

4.9.8 *Exchanger differential pressure resistance.* The heater shall be operated at a fuel pressure within the specified range at optimum fuel-air ratio and with a ventilating air total differential pressure reduced to give a heater ventilating air outlet temperature of 121.1° ±5.6° C (250° ±10° F). The ventilating air outlet pressure shall be at sea-level pressure and the combustion air outlet pressure shall be at least 10 pounds per square inch lower than the ventilating air outlet pressure. At the completion of 4 hours of operation at this condition the heater shall not have been damaged and the iso-thermal ventilating and combustion air flow shall not have changed more than 10 percent at any comparative condition.

4.9.9 *Radio interference.* The heater shall be electrically connected to the specified power source and tested for conducted and radiated radio interference.

4.9.9.1 *Conducted interference.* The equipment and test method used shall conform to Specification MIL–I–6181. The conducted radio interference shall not exceed 200 microvolts in the frequency range 0.15 to 0.20 megacycles and 50 microvolts in the frequency range 0.21 to 20.0 megacycles.

4.9.9.2 *Radiated interference.* The equipment and test method shall conform to Specification MIL–I–6181. The radiated radio intereference shall be less than 2.5 equivalent input microvolts in the range 0.20 to 0.50 megacycles and one equivalent input microvolt in the range 0.51 to 18 megacycles. In the range 100 to 155 megacycles, the noise shall not exceed the receiver background noise.

4.9.10 *Altitude acceptance calibration.* The heater shall be installed in a test chamber connected to a suitable source of ram air of variable pressure and temperature and the output at operating altitudes from sea level to the maximum operating altitude shall be determined. The performance conditions specified by the airframe manufac-

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turer's curve for the specific aircraft in which the heater is to be used shall be established. After the heater has been placed into operation, not more than 5 minutes shall elapse after ignition before the heat output is in accordance with the performance requirements of the same applicable curve. The efficiency shall be determined by using the high heat value of the fuel. The temperature rise of the ventilating air shall not exceed 204° C (400° F), and the exhaust gases 538° C (1000° F).

4.9.11 *Cold starting.* All cold starting tests shall be conducted with ambient air, ventilating and combustion air, and fuel (other than Grade II referee fuel) at a temperature of −54° C (−65° F). When Grade II referee fuel is used, the fuel supplied to the heater shall be at a temperature corresponding to a fuel viscosity of not less than 12 centistokes. The operating voltage for all cold starting tests shall be 17.0 volts d-c. The ignition time as defined in 3.9 shall apply.

4.9.11.1 *Routine cold starting.* The heater shall be maintained at the test conditions for at least ½ hour before each attempt to start. Three consecutive starts shall be accomplished at each test point specified on the applicable curve for the specific aircraft in which the heater is to be used. A maximum of four starting attempts shall be permitted at each test point.

4.9.11.2 *Prolonged cold starting.* The heater shall be maintained at the test conditions for a minimum of 8 hours. A start shall then be attempted at the maximum altitude and maximum differential total pressure at that altitude in accordance with the applicable curve for the specific aircraft in which the heater is to be used. The heater shall ignite satisfactorily. A maximum of two starting attempts shall be permitted.

4.9.12 *Delayed ignition.* The heater shall be operated at the minimum differential total pressures (ventilation and combustion) specified on the applicable curve for the specific aircraft in which the heater is to be used to result in a 100 percent rated output. After the heater has been operated for 5 minutes, it shall be momentarily turned OFF for a period not exceeding 15 seconds after the fire is extinguished and then turned ON again, but with the ignition circuit open so that it does not operate. The heater shall be subjected to this condition with the specified differential total pressure for 5 minutes, at which time the ignition circuit shall be closed. The heater shall start operating with no damage to the heater from any action incident to ignition.

4.9.13 *Fuel pressure variation.* The heater shall operate satisfactorily at rated heat output in the fuel pressure supply range specified by the manufacturer but within the limits of 3.4.3.

4.9.14 *Vibration.* The heater shall be installed on the vibration test stand using the same type of mounts intended for securing the items in the aircraft. The test specimen shall be vibrated in accordance with Procedure XII of Specification MIL–E–5272. The heater need not be operating while conducting this test. Any evidence of damage revealed by subsequent operation and examination shall be cause for rejection.

4.9.15 *Explosion test of ignition system accessories.* All accessories which are designed to be vaporproof and are external of the heater or combustion or ventilating ducts shall be fitted with a ⅛-inch pipe-threaded boss through which air pressure shall be applied until an internal pressure of 30 inches of mercury at 16° C (60° F) above barometric pressure is obtained. The air supply shall then be shut off and the pressure in the accessory equipment shall not change, due to leakage, in 24 hours. A variation of 0.1 inch from the original shall be allowed after correcting for temperature and ambient pressure change. The Explosion test shall be waived on accessories that prove to be vapor tight. All other accessories external of the heater or combustion or ventilating ducts shall be subjected to an Explosion test as follows:

a. Accessories with cases which are not designed to prevent flame or explosion propagation of internal

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case explosions shall be subjected to the Explosion Test, Procedure III, of Specification MIL–E–5272 with the exception that the maximum simulated test altitude shall be that altitude specified on the applicable curve for the specific aircraft in which the heater is to be used.

b. Accessories with cases designed as flame and explosion arrestors shall be tested for the property of preventing the propogation of internal case explosions in accordance with the Explosion Test, Procedure IV of Specification MIL–E–5272.

4.9.16 *Humidity.* The heater shall be subjected to the Humidity Test, Procedure I of Specification MIL–E–5272 except that the total test time shall be 120 hours. At the completion of the test the heater shall not be damaged and shall be operative to the same degree of efficiency as prior to this test.

4.9.17 *Sand and dust.* The heater shall be subjected to the Sand and Dust Test, Procedure I of Specification MIL–E–5272. Any evidence of damage revealed by subsequent operation and examination shall be cause for rejection.

4.9.18 *Fungus resistance.* The heater shall be subjected to the Fungus Resistance Test, Procedure I of Specification MIL–E–5272. Deterioration or corrosion of any internal or external components which could in any manner prevent the heater from meeting operational, maintenance, or servicing requirements during service life shall be cause for rejection.

4.9.19 *Salt spray.* The heater shall be subjected to the Salt Spray Test, Procedure I of Specification MIL–E–5272. The time of exposure shall be 50 hours. At the conclusion of the test, the heater shall be capable of satisfactory operation and shall have suffered no damage as a result of the salt spray.

4.9.20 *Performance.* The performance test shall consist of operating the heater at the minimum differential total pressures (ventilation and combustion) where 100 percent rated heat output is required based on the air-frame manufacturer's curve for the specific aircraft in which the heater is to be used. The input fuel pressure, and voltage shall be in the applicable range. Electrical power and heat output shall meet specification requirements. The efficiency using the high heat value of the fuel shall be at least 65 percent.

4.9.21 *Flight test.* Heaters which are procured as replacements for originally installed equipment or heaters which have not been previously subjected to flight tests shall be flight tested in accordance with applicable requirements of Specifications MIL–H–18325 and MIL–T–18607 for Navy Heaters, and; MIL–A–9482 and MIL–T–5842, for Air Force heaters. For these tests the replacement heater shall be installed in the specific model aircraft in which the originally installed heater was used. Prior to flight testing of the replacement heater, a report showing satisfactory completion of all other tests listed in 4.3.2 of this specification shall be submitted and shall have received the approval of the procuring agency. Flight test shall be arranged with the procuring agency.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Heaters shall be individually preserved and packaged in accordance with Specification MIL–P–7936, Level A or C as specified in the contract or order (see 6.3). Level A packaging shall be in accordance with Method II of Specification MIL–P–116 for heaters having accessories attached and Method III of Specification MIL–P–116 for heaters which do not have accessories attached.

5.2 Packing. Heaters packaged as in 5.1 shall be packed in accordance with Specification MIL–P–7936, Level A, B or C, as specified in the contract or order (see 6.3). Exterior containers shall contain identical quantities and as far as practical shall have the minimum cube and tare consistent with the protection required. Containers shall be

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closed and strapped in accordance with the appendix of the container specification.

5.3 **Marking.** In addition to any special markings required by the procurement document, the interior and exterior containers shall be marked in accordance with Specification MIL–P–7936 and Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The combustion type heater covered by this specification is used in aircraft for cabin heating, and wing and empennage (anti-icing) heating or a combination thereof.

6.2 **Flight test.** This specification provides for flight tests (4.9.21).

6.3 **Ordering data.** Procurement documents should include the following:

a. Title, date and number of this specification.
b. Rating of heaters required. (BTU per hour.)
c. The government test activity, designation of preproduction test samples, and instructions concerning the submittal of the test report when applicable (see 4.3.1.1).
d. Selection of applicable levels of preservation and packaging including submethod of preservation required, (see 5.1). Levels of packaging and packing required (see 5.1) and any special marking required (see 5.3).
e. Detail design, applicable fuel and fuel pressure range, fit and performance requirements of the heater for the specific aircraft installation involved.
f. Specific drawing requirements. (See 4.3.1.2).

6.3.1 *Waiver of preproduction tests.* Under conditions specified in 4.3.1.3, preproduction tests may be waived.

6.4 **Additional tests.** The Government reserves the right to conduct any other tests considered necessary to determine conformance with the requirements of this specification or suitability of the heater for service use.

6.5 **Supersession data.** This specification supersedes the requirements of Military Specification MIL–H–5484, dated 16 Jan 1950; and MIL–H–005484A (Wep), dated 13 June 1962, and MIL–H–00584B (Wep), dated 29 April 1963.

6.6 **Marginal indicia.** The outside margins of this specification have been marked to indicate where changes, deletions, or additions from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and Contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION		
ORGANIZATION *(Of submitter)*		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $
MATERIAL PROCURED UNDER A	☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT	

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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